

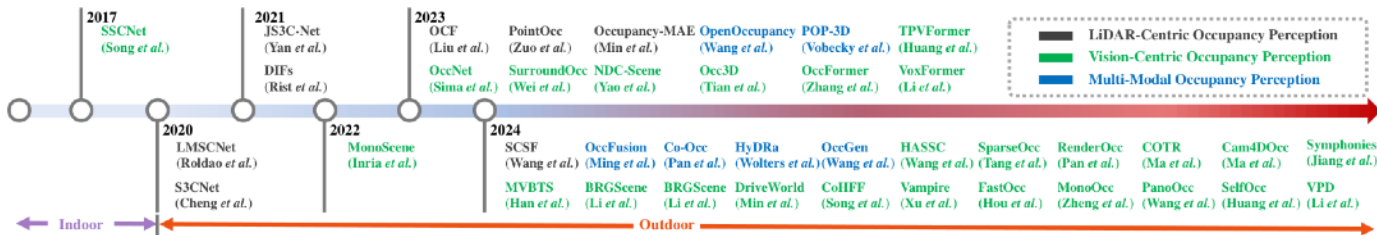
AIX

Autonomous Driving (2)

Mingyu Kim,
May, 19th, 2026

Why Start with CVPR 2022?

- The 2022 Tesla talk is the clearest public moment when Tesla shifts from object lists to occupancy-centric world representation.
- To understand Tesla's later end-to-end story, students should first understand this occupancy pivot.



Occupancy timeline — Tesla's 2022 moment in context

Classical Drivable Space Fails on Unknown Geometry

- The classical drivable-space view works only as long as the world fits predefined categories.
- Tesla's motivation is that unknown objects and irregular geometry still matter for safety.



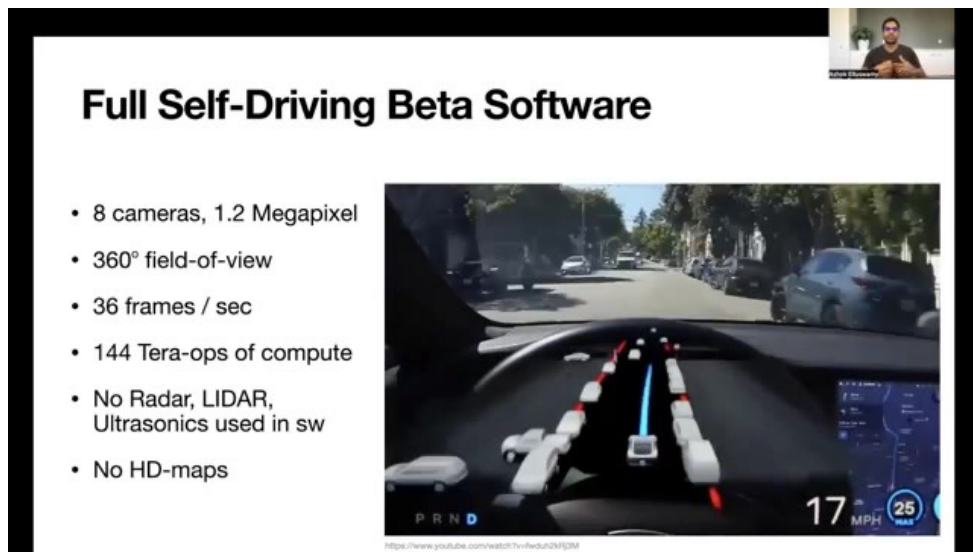
Occupancy Networks: The Motivation in One Slide

- Occupancy is presented as a geometry-first answer to the weaknesses of box-based perception.
- The key message is that dense 3D scene structure matters more than ontology alone.



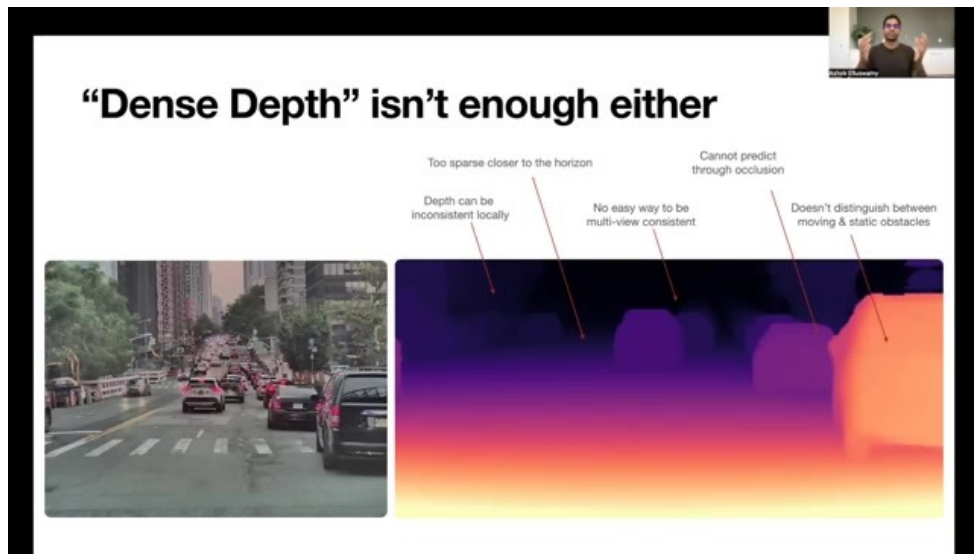
Voxel Intuition: Predict Space, Not Only Classes

- Occupancy turns the scene into a grid of 3D cells that can be reasoned about directly.
- This is the right intuition to teach before any architecture details.



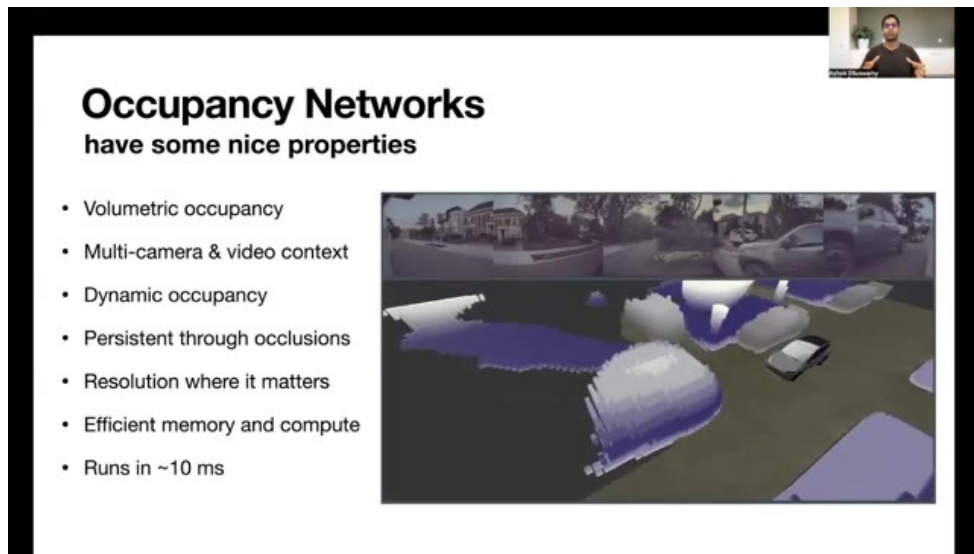
2022 Architecture Overview

- Tesla's 2022 design already treats multi-camera video as the raw substrate for a learned 3D representation.
- The goal is not a list of objects, but occupancy features that can later be decoded into spatial structure.



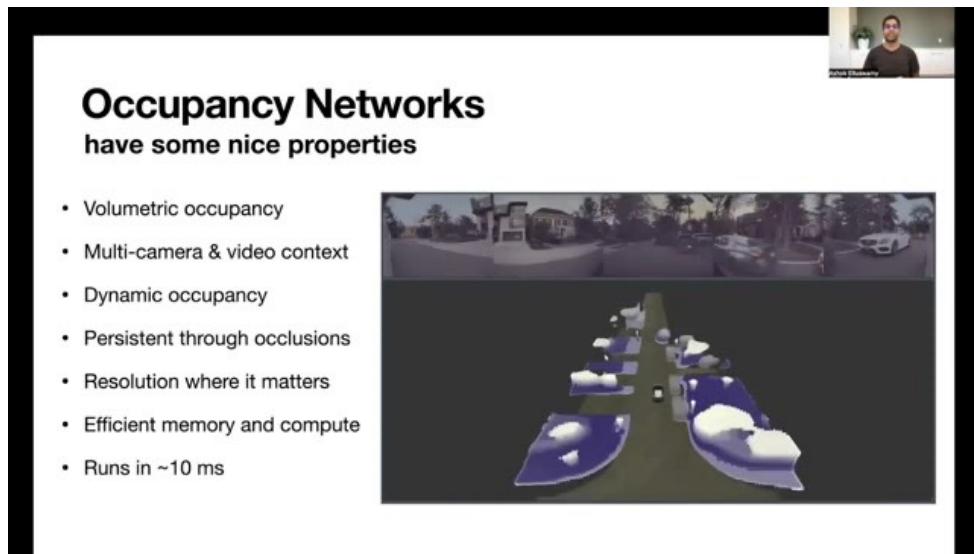
2022 Already Used Transformer-Style Attention

- The 2022 architecture explicitly contains an attention block with Q, K, V and a softmax attention formula.
- So the talk already contains transformer-style attention, even though it is not framed as a pure ViT story.



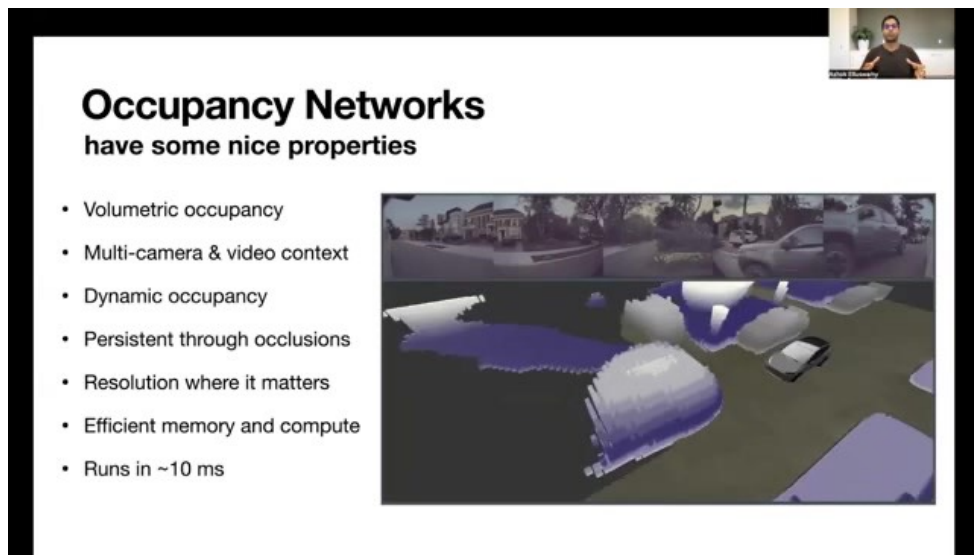
Fixed Queries and Positional Encoding Are Explicit in 2022

- The attention module does not attend blindly; it uses fixed queries and positional encoding.
- This makes the 2022 talk an important bridge between classical AV perception and transformer-era representation learning.



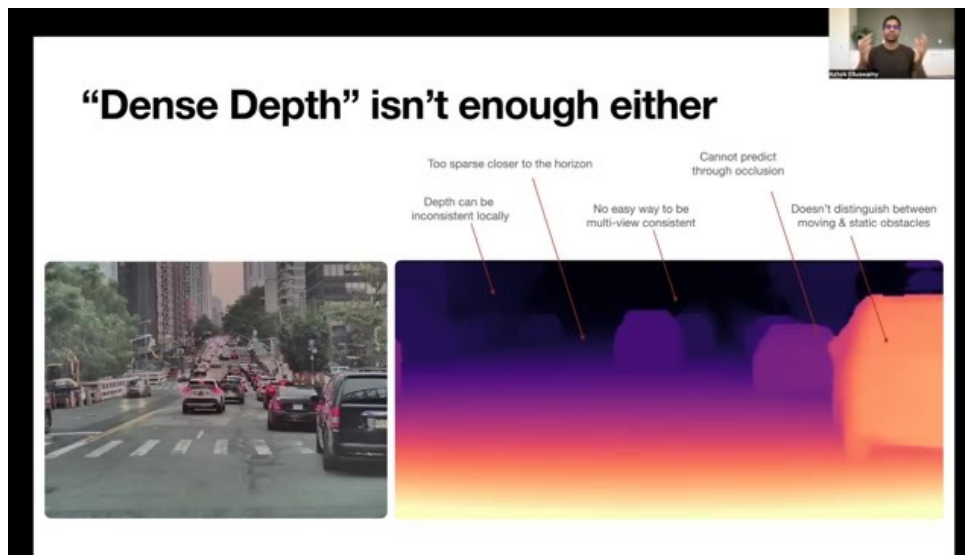
From Occupancy Features to Occupancy Volume

- The architecture does not stop at attention features; it decodes them into a dense occupancy volume.
- This is how a learned latent representation becomes a spatial world model.



The 2022 Talk Is Not a Pure ViT Story

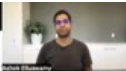
- The backbone shown in the 2022 slide is RegNet + BiFPN, not a pure Vision Transformer encoder.
- The transformer-like component appears in the attention module that lifts image features into occupancy features.



Tesla CVPR 2022: RegNet + BiFPN backbone

Geometry Beats Ontology

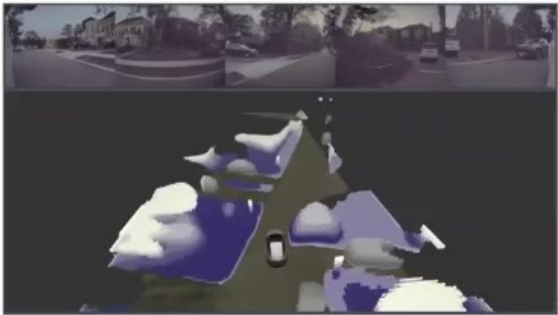
- The talk argues that the world should not be reduced to only named categories.
- Geometry can matter for safe driving even when ontology is missing or wrong.



Occupancy Networks

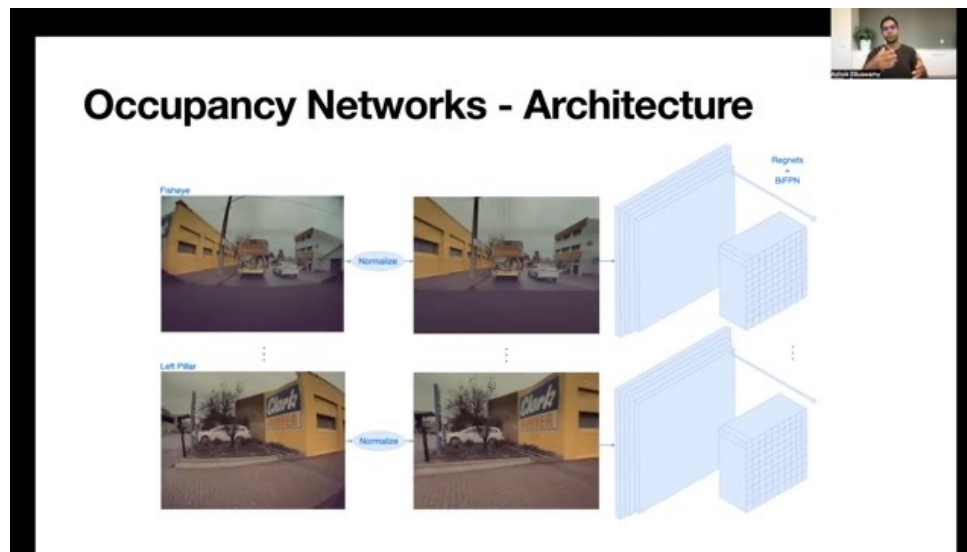
have some nice properties

- Volumetric occupancy
- Multi-camera & video context
- Dynamic occupancy
- Persistent through occlusions
- Resolution where it matters
- Efficient memory and compute
- Runs in ~10 ms



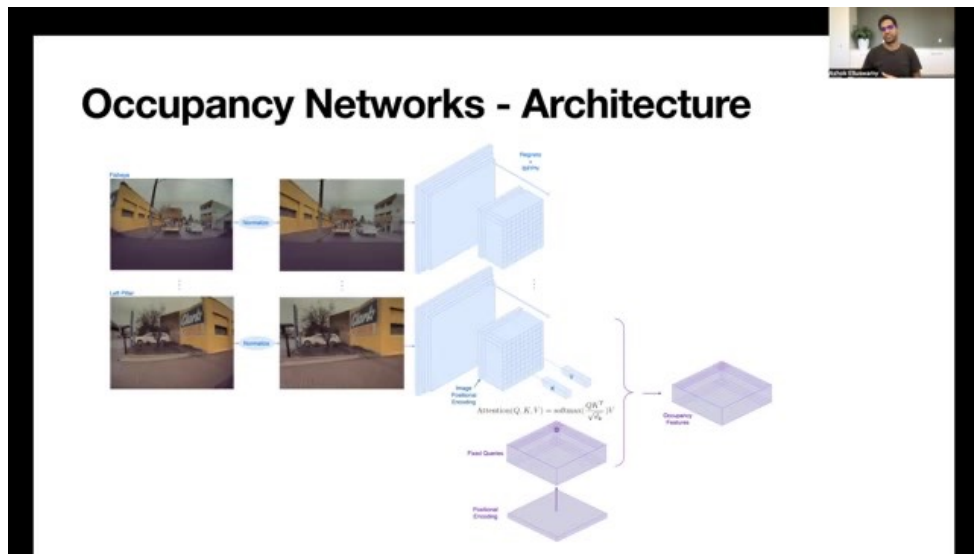
Failure Cases Explain the Ontology Problem

- The ontology story becomes convincing when the slide shows failure cases, not just slogans.
- These examples explain why box detectors are not sufficient as a safety representation.



'False Pedestrians' vs 'True Pedestrians' Is a Driving Problem

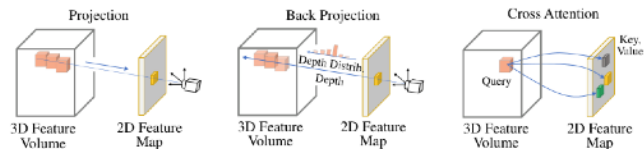
- The point is not only classification accuracy; it is the downstream driving consequence of the mistake.
- This is a very teachable example for second-year students.



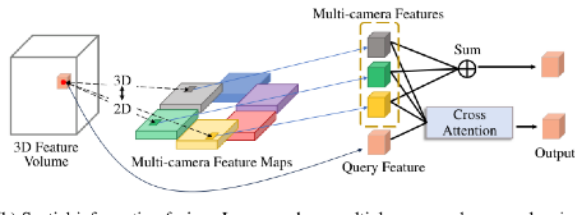
Tesla CVPR 2022: false vs true pedestrians

Why Occupancy Helps Safety and Free-Space Reasoning

- Occupancy can support collision avoidance even when the object class is unknown.
- This is why Tesla framed occupancy as more than a perception upgrade.



a) 2D-to-3D transformation. This serves as the fundamental unit for constructing 3D data from 2D observations, typically by projection [26, 31, 38, 53, 91, 119], back projection [55, 80, 81, 82, 90], and cross attention [4, 36, 76, 113, 118].



Occupancy Is a Unified Scene Representation

- Occupancy joins geometry, free space, and semantics more naturally than stitched task heads do.
- That makes it a better bridge to planning than an object list.

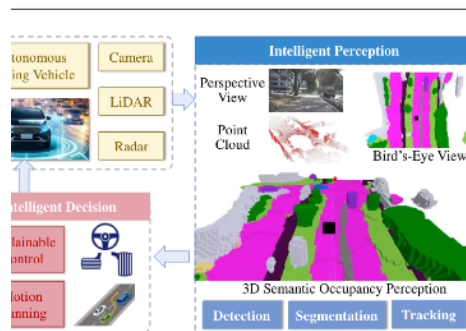


Figure 1: **Autonomous driving vehicle system.** The sensing data from camera, LiDAR, and radar enable the vehicle to intelligently perceive its surroundings. Subsequently, the intelligent decision module generates control and planning driving behavior. Occupancy perception surpasses other perception methods based on perspective view, bird's-eye view, or point clouds, in terms of understanding and density.

In academia and industry, occupancy perception for holistic

2022 Was Representation-Level End-to-End

- The 2022 Tesla story is not yet 'camera directly to steering' in the strongest sense.
- It is best understood as representation-level end-to-end: world state is learned in a unified latent form.



Fig. 1: Survey at A Glance. (a) Pipeline and Methods. We define end-to-end autonomous driving as a learning-based algorithm framework with raw sensor input and planning/control output. We deepdive into 270+ papers and categorize

What 2022 Still Did Not Solve

- Occupancy solves geometry better than boxes, but it does not automatically solve all semantics, temporal reasoning, and transfer.
- This limitation is exactly what opens the door to the 2023 foundation-model story.

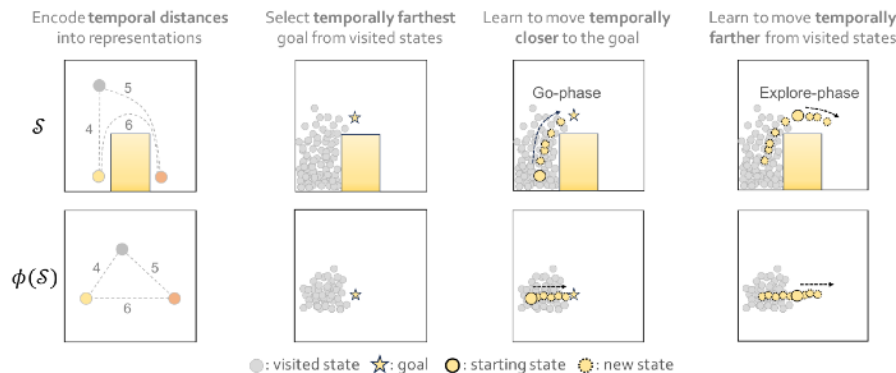
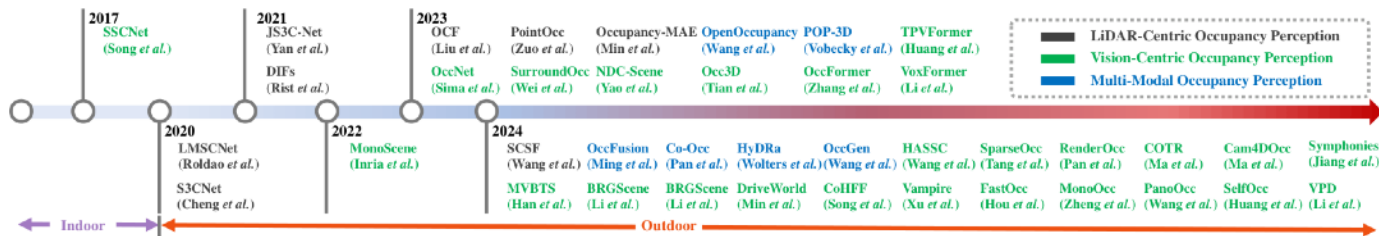


Figure 2: **Overview of TLDR algorithm.** TLDR leverages temporal distance-aware representations for unsupervised GCRL. (a) We start by learning a state encoder $\phi(s)$ that maps states to temporal distance-aware representations. With the temporal distance-aware representations, TLDR (b) selects the *temporally* farthest state from the visited states as an exploratory goal, (c) reaches the chosen goal using a goal-conditioned policy, which learns to minimize temporal distance to the goal, and (d) collects exploratory trajectories using an exploration policy that visits states with large temporal

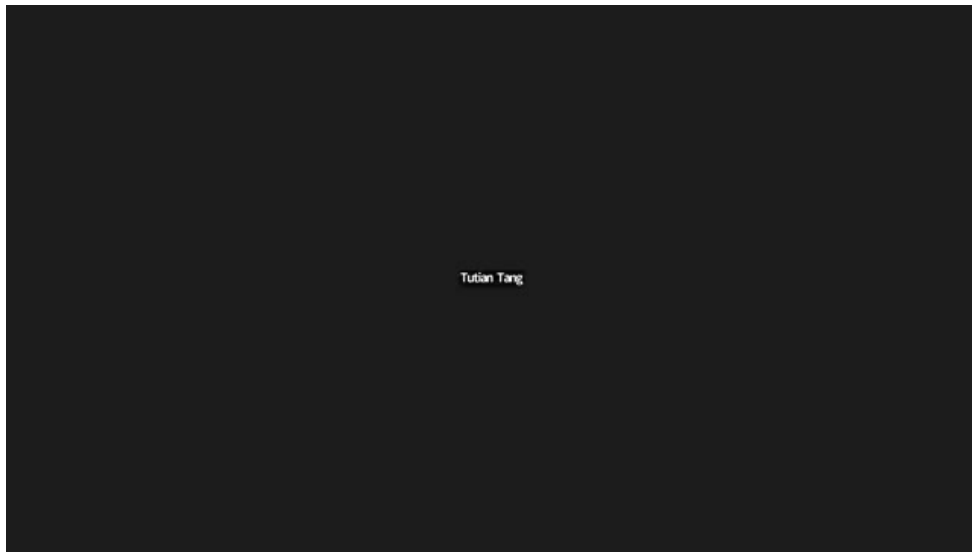
Occupancy Was a Bridge, Not the Final Destination

- Historically, occupancy is a major pivot, but it is also a bridge toward richer world representations.
- That bridge becomes much clearer when we place 2022 back on the occupancy timeline.



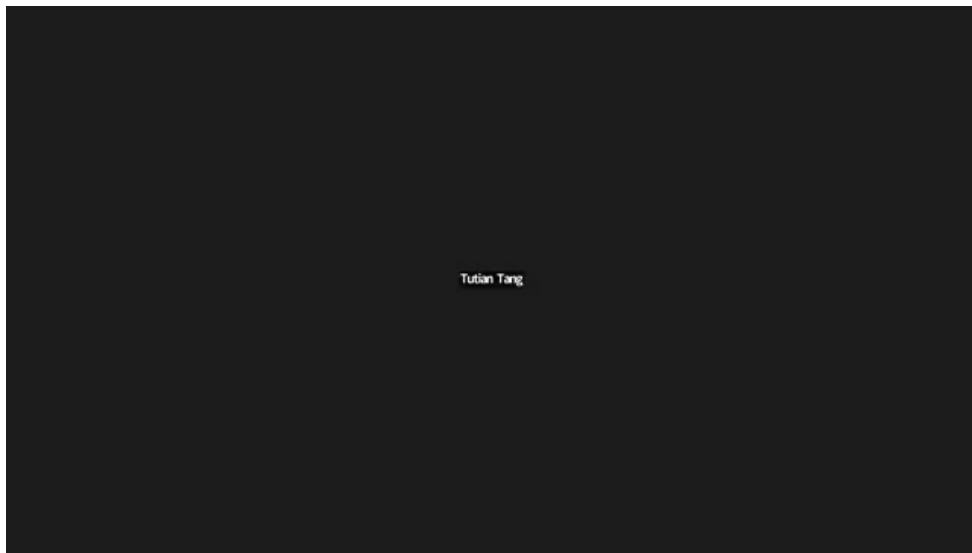
Transition: From Occupancy to Foundation Models

- The next question is no longer only 'how do we represent space?' but 'how do we reuse one learned world representation across many tasks?'
- That is the entry point for the 2023 Tesla talk.



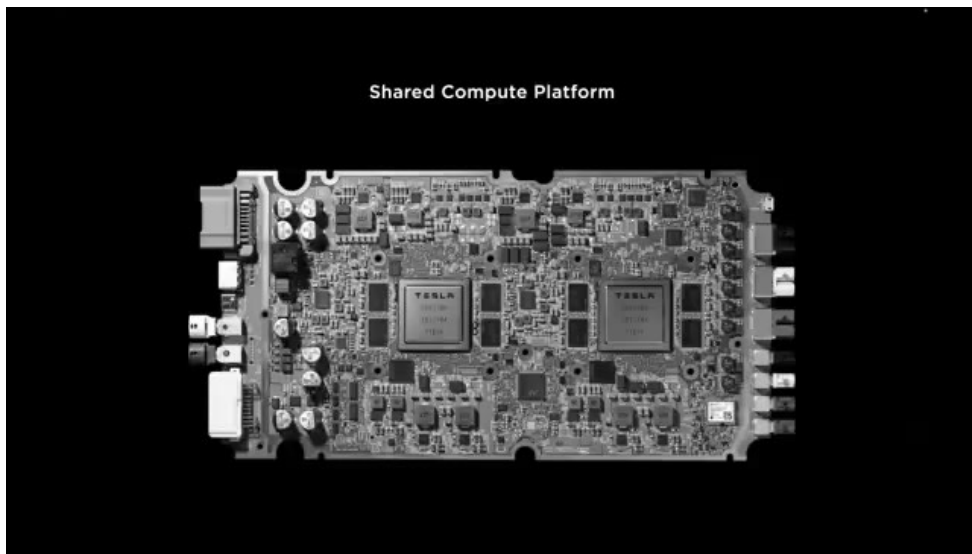
2023 Reframes the Question

- In 2023, the public Tesla story shifts from one representation to a reusable foundation for many driving tasks.
- The move is from a single perception innovation to a broader representation strategy.



2023 Architecture at a Glance

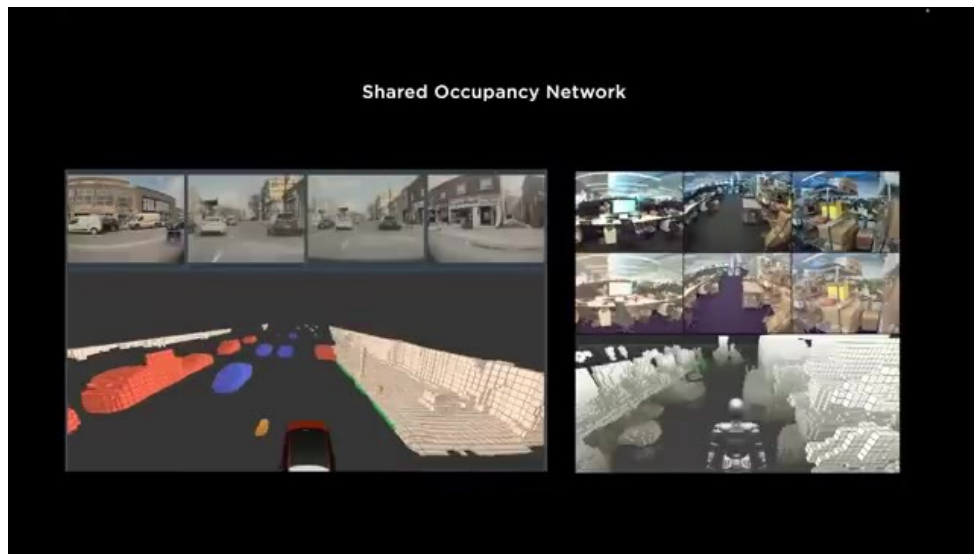
- The 2023 architecture is organized around a shared world representation rather than isolated task heads.
- This is the cleanest single slide for showing Tesla's 'foundation model for driving' idea.



Tesla CVPR 2023: architecture overview

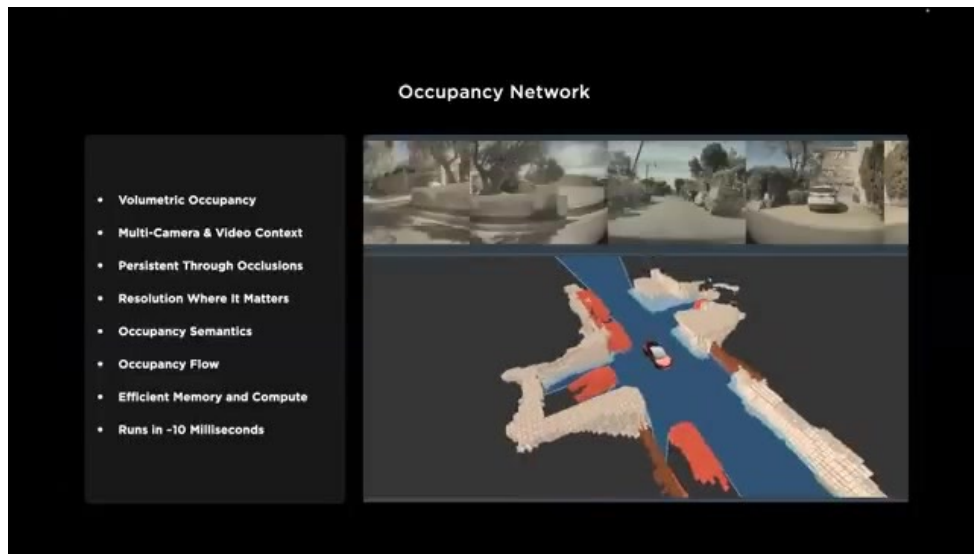
Image Inputs and Feature Backbones

- The model still begins with images, but the goal is no longer 2D-only understanding.
- Feature extraction is the front end of a larger world-model pipeline.



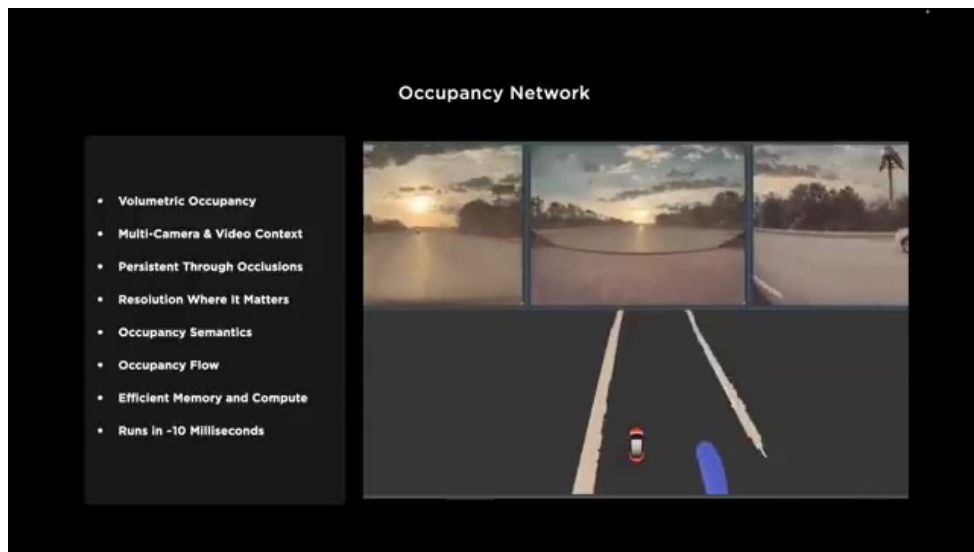
Spatial Attention Becomes Central

- Spatial attention is used to fuse information into a shared driving representation.
- This is one reason the 2023 talk feels closer to a foundation-model architecture than a classic perception stack.



Temporal Alignment Turns 3D Into 4D

- The 2023 story is explicitly temporal: the model should align observations across time, not just across views.
- That is what makes the representation closer to a world model than a snapshot predictor.



Tesla CVPR 2023: temporal alignment block

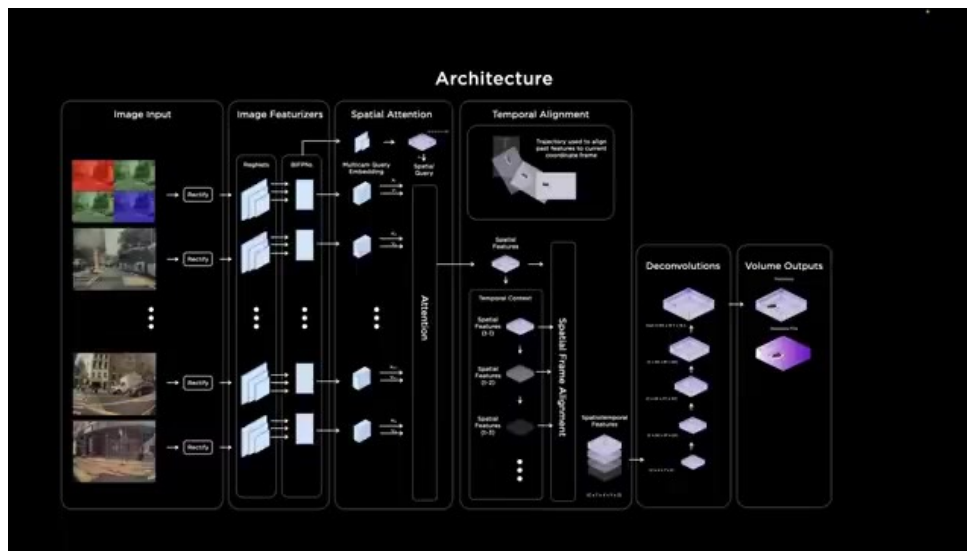
Surface Outputs and Volume Outputs Share One Representation

- Different geometric products are derived from one shared latent representation.
- This is the opposite of training many disconnected perception models.



Queryable Outputs Make Reuse Explicit

- A foundation model matters only if many tasks can query the same latent state.
- This is the clearest architectural cue that Tesla is moving toward shared reuse rather than isolated heads.



Tesla CVPR 2023: task query outputs

Why Foundation Models for Driving?

- The point is not merely to have a larger model, but to have one representation that many downstream tasks can share.
- This mirrors the foundation-model logic already seen in NLP and general computer vision.

Additionally, training a goal-reaching policy to maximize sparse [8] or heuristic [10, 12] goal-reaching rewards is often insufficient for long-horizon goal-reaching behaviors in complex environments.

In this paper, we propose a novel unsupervised GCRL method that leverages **Temporal Distance-aware Representations (TLDR)** to improve both goal-directed exploration and goal-conditioned policy learning. TLDR uses temporal distance (i.e. the minimum number of environment steps between two states) induced by temporal distance-aware representations [1, 13, 14] for (1) selecting faraway goals to initiate exploration, (2) learning an exploration policy that maximizes temporal distance, and (3) learning a goal-conditioned policy that minimizes temporal distance to a goal.

TLDR demonstrates superior state coverage compared to prior unsupervised GCRL and skill discovery methods in complex AntMaze environments, as shown in Figure 1. Our ablation studies confirm that our temporal distance-aware approach enhances both goal-directed exploration and goal-conditioned policy learning. Furthermore, our method outperforms prior work across diverse locomotion environments, underscoring its general applicability.

2 Related Work

Unsupervised goal-conditioned reinforcement learning (GCRL) aims to learn a goal-conditioned

A Typical FM-Enhanced Driving Pipeline

- The survey view makes the Tesla 2023 story easier to interpret because it breaks the system into reusable FM-centered roles.
- This is a good pedagogical bridge between Tesla's talk and the broader research literature.

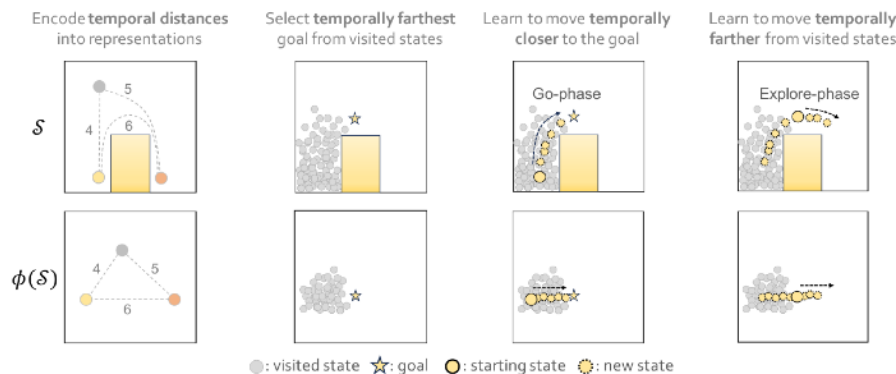


Figure 2: **Overview of TLDR algorithm.** TLDR leverages temporal distance-aware representations for unsupervised GCRL. (a) We start by learning a state encoder $\phi(s)$ that maps states to temporal distance-aware representations. With the temporal distance-aware representations, TLDR (b) selects the *temporally* farthest state from the visited states as an exploratory goal, (c) reaches the chosen goal using a goal-conditioned policy, which learns to minimize temporal distance to the goal, and (d) collects exploratory trajectories using an exploration policy that visits states with large temporal

Specialized Models Fail to Share Context

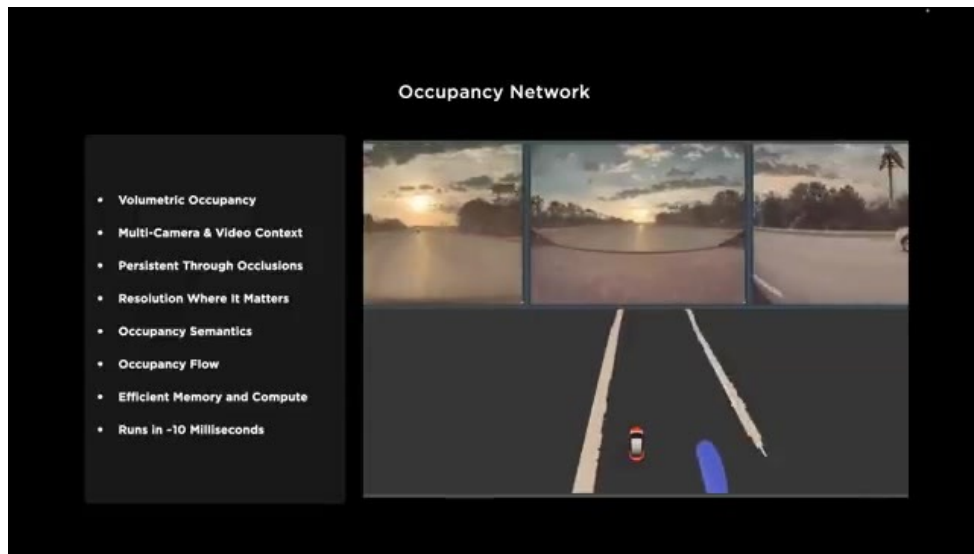
- When detection, mapping, motion forecasting, and planning are trained separately, shared context is lost.
- The foundation-model view tries to reduce that fragmentation.

policy to maximize sparse [8] or heuristic horizon goal-reaching behaviors in complex environments. Supervised GCRL method that leverages supervised learning to improve both goal-directed exploration and distance (i.e. the minimum number of steps to reach the goal) distance-aware representations [1, 12] learning an exploration policy that balances exploitation and exploration. The proposed policy that minimizes temporal difference error compared to prior unsupervised methods in complex environments, as shown in Figure 1. Our approach enhances both goal-directed exploration and distance-aware representations. Furthermore, our method outperforms prior methods in terms of its general applicability.

Reinforcement learning (GCRL) aims to learn a policy that maximizes the expected return over long-term horizons.

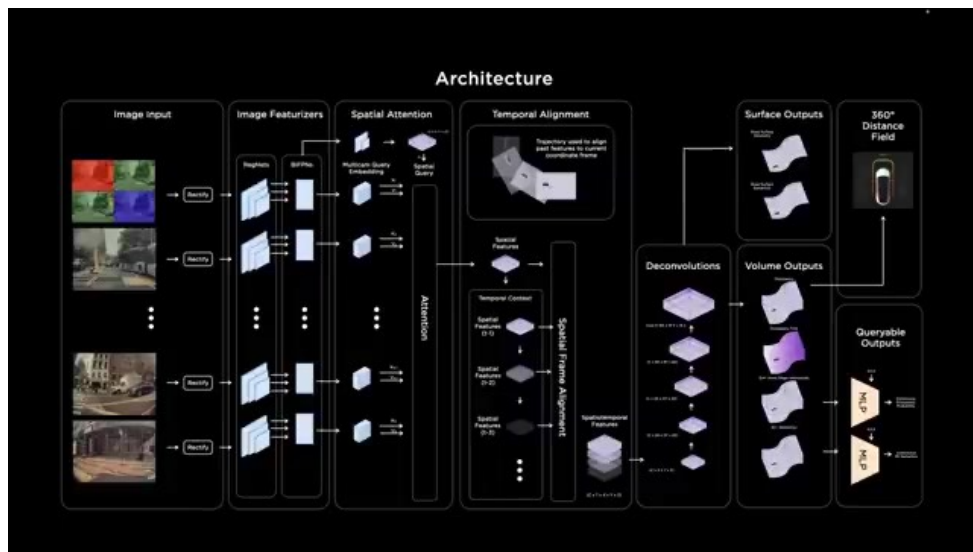
Video Becomes the Natural Training Unit

- A driving model that ignores time is missing the true structure of the problem.
- Video learning improves occlusion recovery, motion reasoning, and future-state consistency.



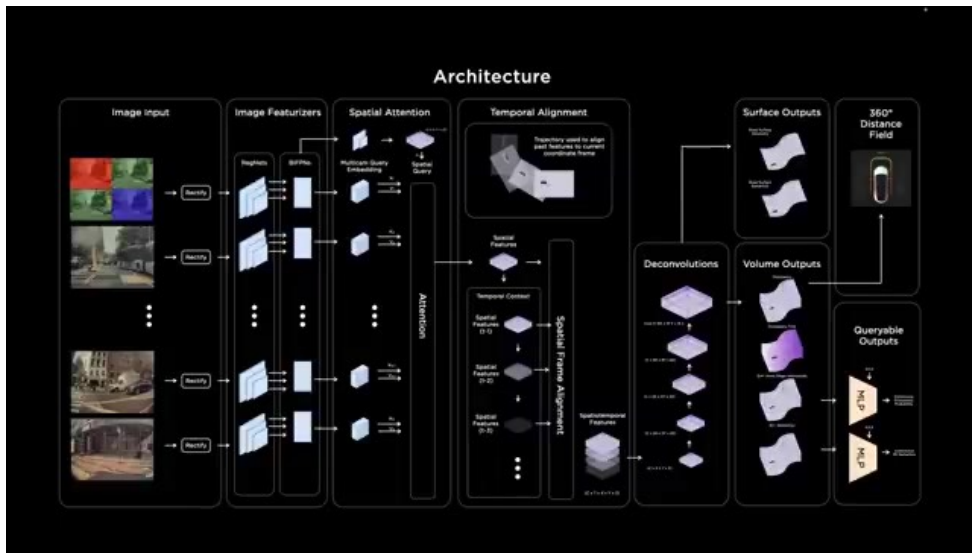
The Data Engine Is Part of the Model Story

- Tesla's data engine is not a side tool; it is part of the modeling strategy.
- Data mining, auto-labeling, retraining, and redeployment jointly define the system.



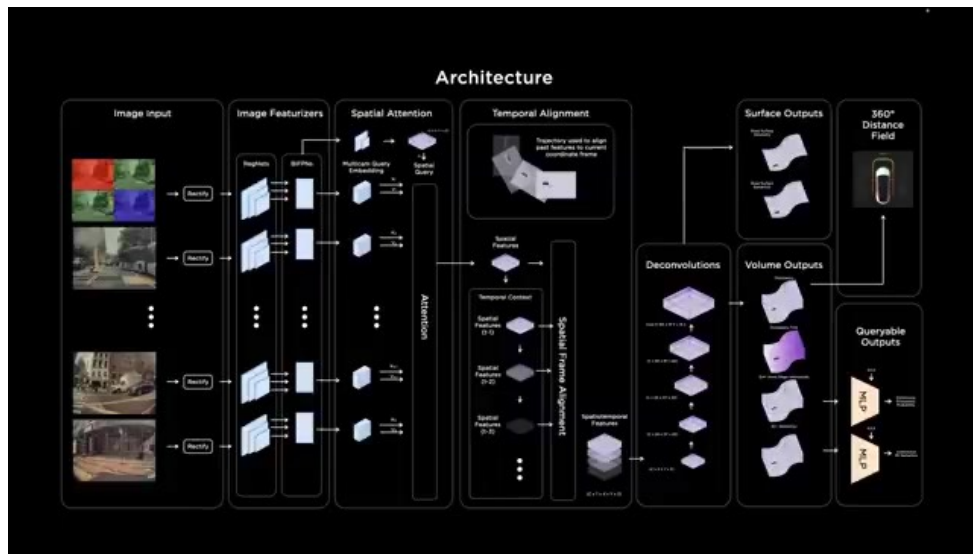
Fleet Learning Closes the Loop

- Foundation-model performance depends on how quickly the system discovers and recycles failure cases.
- This is why Tesla's fleet story matters historically as much as its architecture story.



Auto-Labeling Still Matters in the Foundation-Model Era

- Larger shared models increase, not decrease, the need for scalable supervision.
- Tesla's world-model narrative still depends on auto-labeling and pseudo-labeling.



World Representation Is Upstream of Planning

- A stronger world representation gives the planner richer and more stable context.
- This is why foundation models in driving should be taught as planning enablers, not only perception upgrades.

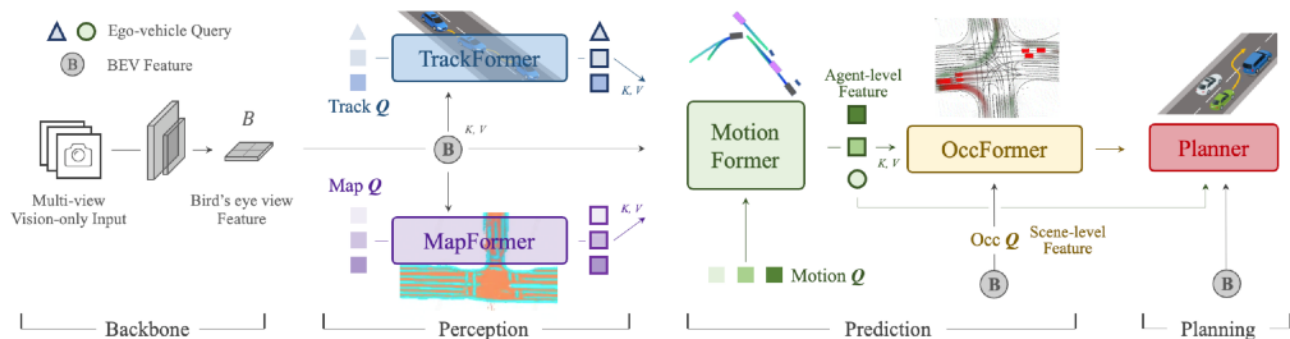


Figure 2. **Pipeline** of Unified Autonomous Driving (UniAD). It is exquisitely devised following planning-oriented philosophy. Instead of a simple stack of tasks, we investigate the effect of each module in perception and prediction, leveraging the benefits of joint optimization from preceding nodes to final planning in the driving scene. All perception and prediction modules are designed in a transformer decoder structure, with task queries as interfaces connecting each node. A simple attention-based planner is in the end to predict future waypoints of the ego-vehicle considering the knowledge extracted from preceding nodes. The map over occupancy is for visual purpose only.

From Occupancy to Reusable Latent State

- Occupancy predicts dense geometry, but the 2023 story tries to turn that into a general reusable latent state.
- That is the conceptual step from occupancy network to vision foundation model.

policy to maximize sparse [8] or heuristic horizon goal-reaching behaviors in complex environments. We propose a supervised GCRL method that leverages occupancy to improve both goal-directed exploration and distance estimation (i.e. the minimum number of steps to reach the goal) by learning distance-aware representations [1, 12]. We also learn an exploration policy that balances exploitation and exploration. Our method outperforms prior methods in terms of exploration efficiency compared to prior unsupervised methods in complex environments, as shown in Figure 1. Our approach enhances both goal-directed exploration and distance estimation. Furthermore, our method outperforms prior methods in terms of its general applicability.

Reinforcement learning (GCRL) aims to learn a policy that maximizes the expected return over long-term horizons.

Why This Supports End-to-End Driving

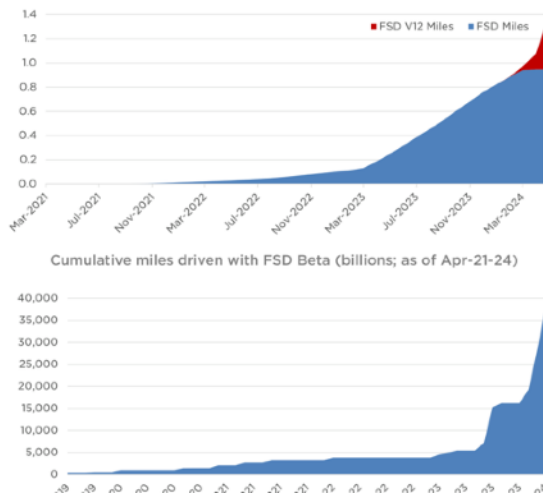
- Once the world representation is shared and rich enough, the planner can depend less on manually stitched intermediate interfaces.
- This is how foundation models connect back to end-to-end driving.



Fig. 1: Survey at A Glance. (a) Pipeline and Methods. We define end-to-end autonomous driving as a learning-based algorithm framework with raw sensor input and planning/control output. We deepdive into 270+ papers and categorize

Scale Means Data + Compute + Deployment

- A foundation model becomes meaningful only when scale reaches data collection, labeling, compute, and deployment together.
- This is also why Tesla's internal infrastructure story matters so much.



Tesla Q1 2024: FSD miles and AI training capacity

The 2023 Story Connects Directly to the 2024 Product Direction

- Tesla's 2023 world-model / foundation-model story is not separate from product deployment; it leads directly into the 2024 end-to-end narrative.
- This is why the 2022 and 2023 talks should be read together with the later official update.

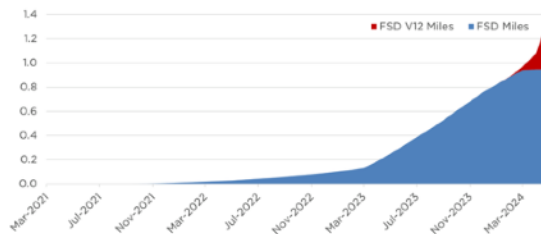
Artificial Intelligence Software and Hardware

We have been investing in the hardware and software ecosystems necessary to achieve vehicle autonomy and a ride-hailing service. We believe a scalable and profitable autonomy business can be realized through a vision-only architecture with end-to-end neural networks, trained on billions of miles of real-world data. Since the launch of FSD (Supervised) V12 earlier this year, it has become clear that this architecture long pursued by Tesla is the right solution for scalable autonomy. To further improve our end-to-end training capability, we will continue to increase our core AI infrastructure capacity in the coming months. In Q1, we completed the transition to Hardware 4.0, our latest in-vehicle computer with increased inference processing power and improved cameras.

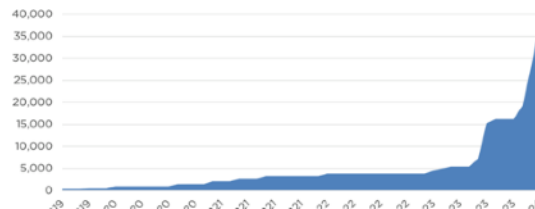
Vehicle and Other Software

We rolled out FSD (Supervised) with a 30-day free trial to eligible cars in U.S. and Canada. FSD (Supervised) can change lanes, select forks to follow routes, navigate around vehicles and objects and make turns. We released Autopark to existing eligible FSD (Supervised) and Enhanced Autopilot customers. We are currently working on ride-hailing functionality that will be available in the future. We believe the Tesla software experience is best-in-class across all our products, and plan to seamlessly layer ride-hailing into the Tesla App.

Battery, Powertrain & Manufacturing



Cumulative miles driven with FSD Beta (billions; as of Apr-21-24)



Tesla Q1 2024: end-to-end statement + charts

2022 vs 2023 Side by Side

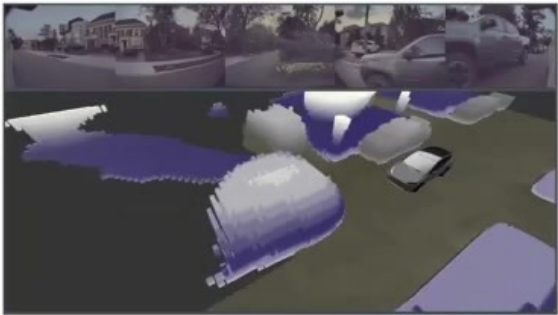
- The 2022 talk centers on occupancy as a new representation, while the 2023 talk centers on one shared representation serving many tasks.
- Put differently: 2022 changes the world state, and 2023 changes how many things that world state can support.



Occupancy Networks

have some nice properties

- Volumetric occupancy
- Multi-camera & video context
- Dynamic occupancy
- Persistent through occlusions
- Resolution where it matters
- Efficient memory and compute
- Runs in ~10 ms



2022 attention/world-representation vs 2023 shared-representation

Historical Synthesis: Modular Stack -> Occupancy -> Foundation Model

- The historical line is not just 'more neural networks'; it is a sequence of changing representations.
- Modular AD breaks the problem into outputs, occupancy learns a denser world state, and foundation models try to reuse that state across tasks.

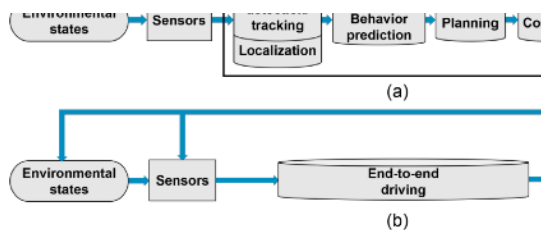


FIGURE 2: Information flow diagrams of: (a) a generic modular system, and

Core functions of a modular ADS can be summarized as: localization and mapping, perception, assessment, planning and decision making, vehicle control, and human-machine interface. Typical pipelines [15], [47], [51]–[56], [56]–[58] start with feeding raw sensor inputs to localization and object detection modules, followed by scene prediction and decision making. Finally, motor commands are generated at the end of the stream by the control module [31], [68].

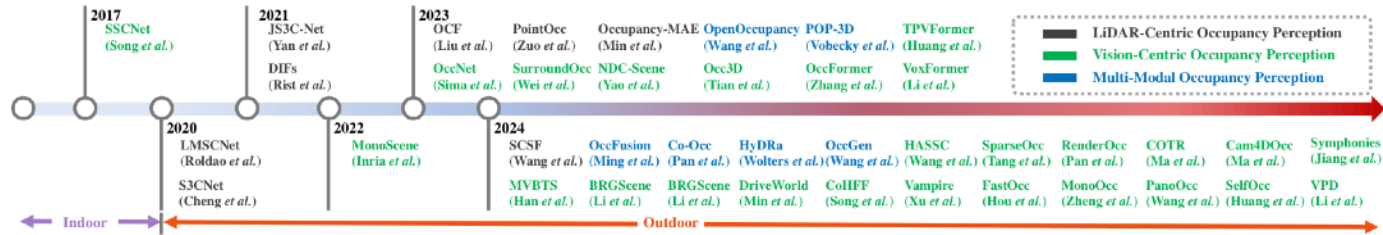
Developing individual modules separately divides the challenging task of automated driving into an easier-to-solve

TABLE 2: Comparison of related works

Related works	Learning
[59]–[63]	Direct
[64], [65]	Deep reinforcement learning

Key Takeaways

- CVPR 2022 is best understood as Tesla's occupancy pivot: geometry-first, dense, and representation-centric.
- CVPR 2023 extends that story into a shared world representation that behaves like a vision foundation model for driving



Timeline figure — final wrap-up